



Accreditation Criteria for Computing Programs

Developed according to the
Graduate Attribute Exemplars of
the Seoul Accord

Version 2025



Common Criteria

Discipline Criteria

Indonesian Accreditation Board for Engineering Education

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Document Control

The International Common Criteria and Criteria Guide version 2025 for Computing Programs have been approved by IABEE Executive Committee on 7 March 2025. This version was based on the 2022 version, which was reorganized in November 2024. Redundant sections have been reduced and its writing has been refined to enhance clarity regarding the fulfillment of the criteria. The 2025 version also welcomes the addition of Cyber Security and similarly-named Computing Programs as well as Data Science and similarly-named Computing Programs to the Discipline Criteria.

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Common Criteria & Criteria Guide

0. Preamble

The Indonesian Accreditation Board for Engineering Education (IABEE) builds this set of Criteria using outcome-based education approach. All computing programs seeking international accreditation from IABEE shall fulfill the following Criteria.

- 0.1. IABEE Common Criteria (CC) and Criteria Guide (CG) are established as a framework to perform accreditation of higher education programs. These CC and CG comprise of elements that shall be fulfilled by the Program to be accredited.
- 0.2. Common Criteria consist of 4 main criteria, following the management approach of PDCA (Plan-Do- Check-Act) continual improvement cycle. Criterion 1 describes the orientation of professional profile and graduate competence, Criterion 2 explains the learning implementation, Criterion 3 explains the assessment of the expected Learning Outcomes, and Criterion 4 explains the continual quality improvements.
- 0.3. P-D-C-A cycle is a continued process for quality improvement. In the Plan step, a way to effect improvement is developed. In the Do step, the plan is carried out. In the Check step, a study takes place between what was predicted and what was observed in the previous step. In the Act step, action is taken on the causal system to effect the desired change.
- 0.4. Programs to be accredited are four-year bachelor of computing programs or other higher education programs which IABEE considered as equivalent, such as Computer Science, Cyber Security, Data Science, Informatics, Software Engineering, Information Systems, Computer Systems, Information Technology or similarly named programs.
- 0.5. The Program is not restricted to single Programs operated by a Department or Faculty. A Program may be formed and/or operated by multiple Departments / Faculties. Programs may include matriculated learning activities outside of its home campus, in conjunction with other higher education institutions.

- 0.6. In cases where multiple Programs of the same nomenclature are offered in multiple locations by the same Program-Operating Institution (such as Programs established according to the *Program Studi di Luar Kampus Utama* (PSDKU) scheme as defined by the Indonesian Ministerial Regulation of *Peraturan Menteri Riset, Teknologi, dan Pendidikan Tinggi* No. 1/ 2017), evaluation by IABEE shall treat the parallel Programs as separate entities.
- 0.7. In cases where the program has multiple streams (such as regular, international, or path-transfer classes), Program Operating-Institution shall explicitly mention the scope for which the evaluation of accreditation is requested. In addition, Program Operating-Institution must be able to make a clear distinction among the streams with regards to permanent records of the graduates, such as certificate and academic transcript.
- 0.8. The Program should promote self-reliance, welfare, advancement, fairness, and justice for the national and global community in general, based on science, technology, culture and sustainable utilization of natural resources.
- 0.9. In addition to the Common Criteria and Criteria Guide, Program seeking for accreditation shall fulfill the Discipline Criteria, eligibility requirements, and other accreditation policies stipulated in the Rules and Procedures of Evaluation and Accreditation (RPEA).

I. Orientation of the Professional Profile and Graduate Competence

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| <p>1.1. The Program shall establish the Autonomous Professional Profile (PPM) to be envisaged as its educational objective, which is based on the internal and external context, taking into account resources, stakeholder input, local and national needs and interests.</p> <hr/> | <p>1.1.1. The Program is required to define the Autonomous Professional Profile (PPM) intended to foster as its educational objectives based on internal and external context, for example vision, mission and values, strategic direction, performance and maturity, resources, relevance, economic and technological trends, standards and regulations, and globalization.</p> <p>1.1.2. The PPM shall describe the professional qualifications and attributes the Program envisages for its graduates to achieve in the early years of their professional career.</p> |
| <p>1.2. Based on the envisaged PPM, the Program shall determine the Graduate Learning Outcomes (CPL) that shall be mastered by students upon completion of their studies. The CPL shall cover aspects of knowledge, skills, and attitudes with the level of depth and breadth as described in the graduate competencies in the following items (a) to (e), as well as additional competencies required by Discipline Criteria (item f).</p> <hr/> | <p>1.2.1. The Program shall establish its own Graduate Learning Outcomes (CPL) based on the PPM to be realized. The CPL shall be mastered by students upon completion of their studies.</p> <p>1.2.2. The CPL shall cover aspects of knowledge, skills, and attitudes with the level of depth and breadth as described in the graduate competencies in the following articles of 1.2.a to 1.2.e</p> <p>1.2.3. The CPL established by the Program shall cover additional outcomes required by the Discipline Criteria relevant to the Program (item f)</p> |

1.2.a. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions with an awareness of the development of trans-disciplinary sciences.

- 1.2.a.1. Ability to analyze a complex computing problem which involves many components, and/or high risks and critical systems.
- 1.2.a.2. Principles of computing include computation, communication, and design.
- 1.2.a.3. Ability to apply other relevant disciplines such as organization processes, finance, medical and engineering.

1.2.b. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.

- 1.2.b.1. Ability to apply information science and technology to the problem-solving process..
- 1.2.b.2. Ability to respond to requirements of the society by utilizing various information sciences and technologies.
- 1.2.b.3. Design ability is composed of items such as complex problem analysis, modelization, extract and define requirements and design, implement and evaluate systems, processes, components and programs.
- 1.2.b.4. Ability to create, select, or adapt and then apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

1.2.c. Communicate effectively in a variety of professional contexts.

- 1.2.c.1. This competence indicates the need of active and effective communication skills; sociocultural perspective should be considered for the acceptability and workability of the implementation of computing works.
- 1.2.c.2. These oral and written communications should include the use of computing standards.
- 1.2.c.3. Ability to communicate oral and/or written internationally recognized languages.

1.2.d. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.	1.2.d.1. Professional responsibilities include relationship among technologies, society and nature, and understanding of health, safety and cultural issues within local and global contexts. 1.2.d.2. Sufficient understanding on issues related to legal principles, such as copyright shall be particularly considered. 1.2.d.3. Ability to comprehend issues related to ethical principles, such as integrity and independence in professional judgment. 1.2.d.4. Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.
1.2.e. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.	1.2.e.1. Ability to collaboratively work with others including people from different fields. 1.2.e.2. Ability to precisely judge and conduct own work during collaborative work. 1.2.e.3. Ability to appropriately judge what others should do and to address to others during collaborative work.
1.3. The Program shall publish the PPM and CPL to the public and shall establish policies and procedures for periodic review and follow up on them consistently.	1.3.1. The Program effectively disseminates PPM and CPL to prospective students, students, faculty, and the general public. 1.3.2. The Program implements policies and procedures for reviewing PPM and CPL, including review inputs, time intervals, and stakeholder involvement. 1.3.3. The Program records and maintain the input, process, output and follow-up actions of the review in a documented system.

2. Learning Implementation

2.1. Curriculum

The Program shall ensure that the academic curriculum is designed to cover the areas of study as mentioned in 2.1.1 points (a)-(c) and the learning process is implemented consistently to lead students to achieve CPL. The curriculum is adequately communicated to faculty members and students. The curriculum design and evidence of its implementation are recorded and maintained in a documented system

2.1.1. Curriculum of the Program shall include the following subject areas:

- (a) Mathematics,
 - (b) Fundamental and advanced topics in the computing discipline,
 - (c) General education, which includes studies of morals, ethics, socio-culture, the environment and management,
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2.1.1.1. Program shall ensure that the curriculum meets the above mentioned subject areas appropriate to computing regardless the subject/course names. The program must ensure that the curriculum devotes adequate attention and time to each component, consistent with the Learning Outcomes, which include mathematics that are appropriate to the discipline, a maximum of 30% general education components, and at least 50% of fundamental and advanced topics in the computing discipline that provide both breadth and depth. The computing topics must include:

- Techniques, skills, and tools necessary for computing practice.
- Principles and practices for secure computing or cyber security.
- Local and global impacts of computing solutions on individuals, organizations, and society

2.1.2. Curriculum shall indicate the structural relationship, contribution and roadmap of each course in building the achievement of CPL during study period. Planning of learning process, course content, organization, learning methods and delivery, as well as assessment methods and criteria shall be established.

2.1.2.1. The curriculum document shall describe the structural relationship showing alignment, roadmap, and contribution of each course in building the achievement of CPL during study period, planning of learning process, the depth and breadth of course content and learning outcomes (CPMK), organization, learning methods and delivery, as well as assessment methods and criteria.

2.1.2.2. The Program shall ensure that the content depth and breadth and specific requirements in each area of curricular study outlined in the Discipline Criteria have been met.

2.1.2.3. The curriculum shall provide students with opportunity to acquire practical experience in implementing the knowledge and skills in an actual working environment, for example in the form of internship or work immersion.

2.1.2.4. The curriculum shall provide students with experience to integrate the knowledge, skills, and attitudes acquired during previous learning processes in a major computing project as culmination of the curriculum (capstone project), in which the students demonstrate their ability to develop solutions to complex computing problems. This project should have a significant credit load and function as a means of demonstrating the fulfillment of many CPL.

2.1.3. The implementation of the learning process, assessment of course learning outcomes (CPMK) achievement, evaluation and follow-up to improve the quality of learning shall be recorded and maintained in a documented system to ensure consistent and controlled implementation of the curriculum.

2.1.3.1. The Program is required to implement educational activities for students to achieve CPL, which include learning process, assessment of course learning outcomes (CPMK) achievement, evaluation and follow-up to improve the quality of learning.

2.1.3.2. Assessment is a systematic process of observing learning and rating of student performance against the learning expectations. Assessment of learning outcomes is the process of appraising knowledge, know-how, skills, and/or competencies of an individual against predefined criteria (learning expectations).

2.1.4. The Program shall communicate the curriculum to faculty and students and shall establish policies and procedures of periodic review and implementing it consistently

2.1.3.3. Evidence may take the forms of Semester Study Plan (RPS) and its realization, course portfolio, assessment instruments, artefacts of student works, evaluation meeting minutes, and other relevant documents of follow-up actions.

2.1.4.1. The Program shall communicate the curriculum effectively to faculty members and students.

2.1.4.2. The Program implements policies and procedure of periodical curriculum review, which covers review input, process, and output, including stakeholder involvement.

2.1.4.3. The input, process and output of curriculum reviews, as well as its follow-up action is recorded and maintained in a documented system.

2.2. Faculty

The Program shall be supported by an adequate number, qualifications and competencies of faculty to carry out academic activities. The Program shall ensure faculty members' role and contribution, facilitate communication among faculty members, and utilize faculty member's academic experience and competence to support achievement of CPL.

2.2.1. The Program shall provide adequate number, qualifications, and competencies of faculty to serve academic and supporting activities.

2.2.1.1. The Program shall describe qualifications of the faculty and their adequacy to cover all curricular areas and to meet the criteria.

2.2.1.2. This description shall include the composition, size, experience and the extent and quality of faculty member involvement in interactions with students, student advising, and oversight of the Program.

2.2.1.3. The Program shall ensure that its faculty members have opportunities to develop their professional competencies through activities such as sabbaticals leave, trainings, workshops, seminars, industrial internship, community services, etc.

2.2.2. The Program ensures that the faculty play a role and contribute to the achievement of CPL through planning, implementation, course evaluation and improvement, as well as student guidance and other forms of contribution

2.2.2.1. The Program shall describe the role played by the faculty with respect to the course creation, modification, and evaluation, and with respect to the definition, revision, and attainment of the Learning Outcomes.

2.2.2.2. The Program shall evaluate the performance of faculty educational activities.

2.2.3. The Program shall facilitate good communication among faculty members to develop close cooperation in organizing academic activities to improve CPL achievement

2.2.3.1. The Program shall define and set up communication network among faculty members for close collaboration among the courses set in the curriculum to obtain better educational results.

2.2.4. The Program encourages the use of faculty's knowledge and experience in research activities, community service, and other competency development to improve teaching and learning quality

2.2.4.1 The Program should provide evidence with respect to the contribution of faculty's updated knowledge, experience, and competency development are used to enhance student learning, such as course material improvement, delivery method, illustration enrichment, case studies, etc.

2.3. Students and Academic Atmosphere

The Program shall establish a system for accepting students, monitoring study progress, and creating a conducive academic atmosphere

2.3.1. The Program shall have student admission policy and procedure, which include establishing requirements and processes for selecting new students, transfer students, credit transfer, and implementing the policy and procedure consistently.

2.3.1.1. The Program shall have written policies on student admission, covering the requirements and the process for accepting new students, including information on how the Program ensures and documents that the applicants are meeting the prerequisites and how it handles cases where the prerequisite have not been met.

2.3.1.2. The Program shall describe the requirements and process for accepting transfer students and transfer credits.

2.3.2. The Program shall establish and implement a system for monitoring study progress and evaluating student performance and maintain records of the process and results.

2.3.2.1. A system for monitoring student progress and evaluating performance includes student academic advising to support student success. The system shall provide reliable records on student performance to facilitate effective advising and function as early warning for the program to take the necessary efforts in ensuring CPL fulfillment by the students.

2.3.2.2. The Program shall provide the students with advice and counseling services on both academic and non-academic aspects, as well as career guidance. The Program shall describe the process of advising and providing career guidance to students, how often students are advised, and who provides the advising.

2.3.3. The Program shall build and maintain an academic atmosphere that is conducive to successful learning.

2.3.3.1. The Program shall develop supporting activities to create and maintain a good academic atmosphere for learning, such as by providing student guidance and counseling on academic as well as non-academic aspects, encouraging co-curricular activities to build character and soft skills through guest lectures, studium generale, student involvement in faculty's research, and participation in competitive activities and scientific forums.

2.3.1.2. The Program shall describe the requirements and process for accepting transfer students and transfer credits.

2.3.4. The Program shall seek to foster students' entrepreneurial spirit to shape and strengthen their life skills.

2.3.4.1. An entrepreneurial spirit shall be emphasized in the learning process. An entrepreneurial spirit is characterized by, among other things, a strong sense of purpose, perseverance, reasoning power, open-mindedness, professionalism, and a passion for learning.

2.4. Facilities

The Program shall ensure the availability, maintenance, currency, and security of facilities and infrastructure, as well as nurture work health and safety culture to support an effective learning process.

2.4.1. The Program shall ensure the provision of infrastructure and physical facilities used in the learning process and supporting activities to create a conducive academic atmosphere.

2.4.1.1. The Program shall describe the facilities in terms of their ability to support the attainment of the CPL and to provide an atmosphere conducive to learning, for example: offices and their supporting equipment (administration and secretariat, lecturers and assistants), classrooms and their supporting equipment, computing laboratories and facilities, computing resources, libraries, sports facilities, and places of worship.

2.4.2. The Program shall evaluate the provided facilities in terms of adequacy, up-to-datedness, and their accessibility by students to ensure effective learning towards fulfilling CPL.

2.4.2.1. The evaluation shall enable the Program to identify the gap between the existing and the required facilities, and to plan the necessary effort to improve and maintain the effective support to student learning.

2.4.2.2. The evaluation shall cover the quantity, quality, up-to-datedness, and accessibility aspects of the facilities.

2.4.3. The Program shall implement policies and procedure for maintaining and updating of facilities and infrastructure.

2.4.3.1 The Program shall describe the policies and procedures it adopts for maintaining and upgrading the tools, equipment, computing resources, laboratories, library, and other facilities used by students and faculty.

2.4.4. The Program shall ensure that facilities and infrastructure are safe for use and shall strive to create a safety culture and a healthy working environment.

2.4.4.1. The Program shall manage safety, health, and environment to ensure safe and appropriate utilization of tools, equipment, computing resources, laboratories, and other physical facilities.

2.4.4.2. The Program should seek to nurture a safety culture by, for example, creating participatory values, attitudes, behaviors, and competencies for safe actions among the academic community.

2.5. Institutional Responsibility

The Program Operating Institution (Institution) shall be committed to and responsible for ensuring that the Program's educational activities run effectively and are sustainable through a good governance system and the provision of adequate resources and funding

2.5.1. The Institution shall build a governance system that ensures effective involvement of Program leaders in decision-making that impacts quality, sustainability, and fulfilment of these Accreditation Criteria.

2.5.1.1. The Program shall describe the governance of the program and its adequacy to ensure the quality and continuity of the program and how the leadership is involved in decisions that affect the Program.

2.5.2. The Institution shall establish and implement budgeting policy and procedure for implementing Program activities and providing the necessary facilities and infrastructure.

2.5.2.1. The Program shall describe the process used to establish the Program's budget and provide evidence of continuity of institutional support for the Program, including the sources of financial support for both permanent (recurring) and temporary (one-time) funds.

2.5.3. The Institution shall support the provision of supporting staff and the development of their professional competencies.

2.5.3.1. The Program shall describe the adequacy of the supporting staff (administrative, instructional, and technical) in terms of the quality and quantity provided to the Program.

2.5.3.2. The Program shall facilitate professional development activities for supporting staff to improve the quality of education services.

2.5.4. The Institution shall facilitate the Program in developing networks and collaboration with various parties both domestically and abroad to support the quality of education, research, and community service.

2.5.4.1. The Institution shall make efforts to develop partnership with external institutions such as government offices, industry, research centers, and community units to foster the *Tridharma* of higher education institutions (teaching-learning, research, and community engagement).

2.5.4.2. The Program shall strive to improve the student learning process through the engagement of academia, business, and/or the government in the development of local region through the use of local resources.

3. Assessment of the Learning Outcomes

3.1. The Program shall ensure that CPL assessment processes are planned and carried out periodically using an appropriate method for each CPL. The method covers the establishment of performance indicators, assessment techniques, planned schedule, and acceptance criteria.

3.1.1. The Program shall define the performance indicators and acceptance criteria for each of the established CPL as assessment reference and evidence of CPL achievement.

3.1.1.1. Performance indicators refer to the means by which an objective can be judged to have been achieved or not achieved. Indicators are used to observe progress and to measure actual results compared to expected results (acceptance criteria). Indicators are, therefore, tied to learning outcomes and serve as yardsticks, by which to measure its degree of achievement. Performance indicators are quantitative tools and are usually expressed as a rate, ratio, or percentage.

3.1.1.2. Assessment of CPL is one or more processes that identify, collect, and prepare data to evaluate the extent of CPL attainment at program-level.

3.1.2. The Program shall establish appropriate assessment methods and planned schedule of assessment for each CPL.

3.1.2.1. Assessment methods are techniques or instruments to measure CPL attainment. Assessment shall at least apply relevant quantitative direct measures, and preferably strengthened by qualitative indirect measures as appropriate. Appropriate sampling approach may be used as part of Program assessment process.

3.1.2.2. The Program shall define periodic assessment schedule of CPL at an appropriate interval.

3.1.3. The Program shall ensure that the CPL assessment process and results are recorded and maintained in a documented system.

3.1.3.1. The process and results of CPL assessment are recorded and maintained in a documented system in such a way to enable meaningful data analysis.

3.2. The Program shall ensure that each graduate meets all determined CPLs and other graduation requirements.

3.2.1. The Program shall implement policies and mechanisms to ensure each graduate achieves CPL and other graduation requirements.

3.2.1.1. The Program shall establish effective policy and procedures to ensure that its graduates meet all graduation requirements.

3.2.2. Fulfilment of graduation requirements is recorded and maintained in a documented system.

3.2.2.1. The instrument, the process and results of graduation requirement review shall be documented. Records are maintained as evidence that all graduates have been evaluated and that all CPL have been fulfilled.

4. Continual Quality Improvement

4.1. The Program shall ensure that there is a periodic and continual quality improvement process, which is based on the results of measuring the achievement of CPL.

4.1.1. The Program shall conduct periodic analysis and evaluation of CPL assessment, which includes identification of issues, fulfillment against established performance targets, and their root causes.

4.1.1.1. Continual quality improvement means the ongoing improvement of processes that lead to the achievement of higher levels of performance. To ensure this improvement, the Program shall run its educational activities by implementing a quality assurance system follows the P-D-C-A cycle as described in the preamble.

4.1.1.2. Evaluation is one or more processes for interpreting the data and evidence accumulated from assessment processes. Evaluation of the education system is an evaluation of overall achievement of the program performance as a basis for continual quality improvement.

4.1.2. The Program shall utilize the results of the CPL achievement evaluation to make decisions to continually improve quality and performance.

4.1.2.1. The output of the evaluation shall contain recommendations on the improvement of overall Program performance, such as curriculum, learning materials, methods of delivery, learning and assessment methods, as well as suitability and adequacy of the learning outcomes with regards to the needs of stakeholders and resources.

4.2. The Program shall ensure that the quality improvement decisions are implemented, and their effectiveness are evaluated. Evidence of the implementation of quality improvements and effectiveness is recorded and maintained in a documented system that enables relevant parties to have access.

4.2.1. The Program shall implement the quality improvement decisions and evaluate their effectiveness as evidence of consistent implementation of PDCA cycle.

4.2.1.1. Implementing and evaluating the effectiveness of continual improvement decisions are important indicators to show a completed PDCA cycle.

4.2.2. The Program shall maintain a documented system of the implementation of quality improvement decisions and ensure its accessibility.

4.2.2.1. The documentation and records of evaluation process, results, and its follow-up decisions shall be maintained as evidence that the decisions have been conducted. These documentation and records shall be accessible to the relevant parties.

Discipline Criteria

Discipline Criteria for Computer Science, Informatics, and Similarly-named Computing Programs

Lead Society(ies):

- *Asosiasi Pendidikan Tinggi Informatika dan Komputer (APTIKOM)* – Association of Higher Education Programs in Informatics and Computer

These program criteria apply to computing programs using computer science or similar terms in their titles.

Learning Outcomes 1.2.f.

In addition to graduate competencies of 1.2.a through 1.2.e required by the Common Criteria, graduates of the program will also have an ability to apply computer science theory and software development fundamentals to produce computing-based solutions

Curriculum

The curriculum requirements specify topics, but do not prescribe specific courses. These requirements are:

- (a) Computer science: At least 48 SKS (or equivalent) that must include:
 1. Substantial coverage of algorithms and complexity, computer science theory, concepts of programming languages, and software development.
 2. Substantial coverage of at least one general-purpose programming language.
 3. Exposure to computer architecture and organization, information management, networking and communication, operating systems, and parallel and distributed computing.
 4. The study of computing-based systems at varying levels of abstraction.
 5. A major project that requires integration and application of knowledge and skills acquired in earlier course work.
- (b) Mathematics: At least 12 SKS (or equivalent) that must include discrete mathematics and must have mathematical rigor at least equivalent to introductory calculus. The additional mathematics might include course work in areas such as calculus, linear algebra, numerical methods, probability, statistics, or number theory.

Discipline Criteria for Computer Systems and Similarly-named Computing Programs

Lead Society(ies):

- *Asosiasi Pendidikan Tinggi Informatika dan Komputer (APTIKOM)* – Association of Higher Education Programs in Informatics and Computer

These program criteria apply to computing programs using computer system or similar terms in their titles.

Learning Outcomes 1.2.f.

In addition to graduate competencies of 1.2.a through 1.2.e required by the Common Criteria, graduates of the program will also have an ability to design computers, computer-based systems, and networks that include both hardware and software as well as their integration

Curriculum

The curriculum requirements specify topics, but do not prescribe specific courses. These requirements are:

1. Exposure to computer architecture and organization, information management, networking and communication, operating systems, and parallel and distributed computing.
2. The study of computing-based systems at varying levels of abstraction.
3. Computer system administration and system maintenance.
4. Computer system integration and system architecture.
5. Mathematics, including discrete mathematics, probability and statistics.

Discipline Criteria for Cyber Security and Similarly-named Computing Programs

Lead Society(ies):

- *Asosiasi Pendidikan Tinggi Informatika dan Komputer (APTIKOM)* – Association of Higher Education Programs in Informatics and Computer

These program criteria pertain to computing programs that use cybersecurity, cyber operations, computer security, information assurance, information security, computer forensics, or similar terminology in their titles.

Learning Outcomes 1.2.f.

In addition to graduate competencies of 1.2.a through 1.2.e required by the Common Criteria, graduates of the program will also have an ability to implement security concepts and practices to sustain operations amidst risks and threats.

Curriculum

The curriculum requirements are in addition to the General Criteria curriculum requirements and specify topics, but do not prescribe specific courses. The requirements are as follows:

- (a) At least 45 semester credit hours (or equivalent) in computing and cybersecurity coursework. The coursework must encompass:
 1. Implementation of the crosscutting principles of confidentiality, integrity, availability, risk, adversarial reasoning, and systems analysis.
 2. Essential subjects from each of the following:
 - a) Data Security: safeguarding data when stored, processed, and transmitted.
 - b) Software Security: the creation and utilization of software that consistently maintains the security attributes of safeguarded information and systems.
 - c) Component Security: the security considerations involved in the design, procurement, testing, analysis, and maintenance of components incorporated into larger systems.
 - d) Connection Security: the safeguarding of connections between components, encompassing both physical and logical aspects.
 - e) System Security: the security considerations of software-based systems comprised of components and interconnections.
 - f) Human Security: the examination of human conduct concerning data protection, privacy, and threat alleviation.
 - g) Organizational Security: safeguarding entities against cybersecurity threats and controlling risks to facilitate the successful achievement of the companies' objectives.
 - h) Societal Security: facets of cybersecurity that significantly affect society at large.
 3. Advanced cybersecurity subjects that expand upon overarching principles and foundational themes to enhance depth.
- (b) A minimum of 6 semester credit hours (or equivalent) in mathematics, which must encompass discrete mathematics and statistics.

Discipline Criteria for Data Science and Similarly-named Computing Programs

Lead Society(ies):

- *Asosiasi Pendidikan Tinggi Informatika dan Komputer (APTIKOM)* – Association of Higher Education Programs in Informatics and Computer

These program criteria are applicable to computing programs that include the terms "data science," "data analytics," or similar terms in their titles.

Learning Outcomes 1.2.f.

In addition to graduate competencies of 1.2.a through 1.2.e required by the Common Criteria, graduates of the program will also have an ability to apply theory, techniques, and tools throughout the data science lifecycle and apply the resulting knowledge to meet the demands of stakeholders..

Curriculum

The curriculum requirements are in addition to the Common Criteria curriculum requirements and specify topics, but they do not prescribe specific courses.

These prerequisites are as follows:

A minimum of 45 semester credit hours (or the equivalent) of data science coursework that must encompass the following:

1. The fundamental topics of the data science lifecycle:
 - a) Data acquisition and representativeness
 - b) Data management
 - c) Data preparation and integration
 - d) Data analysis
 - e) Model development and deployment
 - f) Visualization and communication of the knowledge derived from the data
2. Concepts that are applicable to the data science lifecycle:
 - a) Data ethics, which encompasses legitimate use and algorithmic fairness
 - b) Governance, which includes privacy, security, and stewardship
 - c) Applied statistical and mathematical topics, including inference, modeling, linear algebra, probability, and optimization
 - d) Computing, which includes data structures and algorithms
3. Coursework in advanced data science that offers sufficient depth.
4. The inclusion of a minimum of one application area to establish a context for data science activities.
5. A significant project that involves the integration and application of knowledge and skills that have been acquired in previous courses and integrates an application area.

Discipline Criteria for Information Systems and Similarly-named Computing Programs

Lead Society(ies):

- *Asosiasi Pendidikan Tinggi Informatika dan Komputer (APTIKOM)* – Association of Higher Education Programs in Informatics and Computer

These program criteria apply to computing programs using information systems or similar terms in their titles

Definition

Information Systems Environment – An information systems environment is an organized domain of activity within which information systems are used to support and enable the goals of the activity. Examples of information systems environments include (but are not limited to) business, health care, government, not-for-profit organizations, and scientific disciplines.

Learning Outcomes 1.2.f.

In addition to graduate competencies of 1.2.a through 1.2.e required by the Common Criteria, graduates of the program will also have an ability to support the delivery, use, and management of information systems within an information systems environment.

Curriculum

The curriculum requirements specify topics, but do not prescribe specific courses. These requirements are:

- (a) Information systems: at least 36 SKS (or equivalent) that include coverage of fundamentals and applied practice in application development; data and information management; information technology infrastructure; systems analysis, design and acquisition; project management; and the role of information systems in organizations.
- (b) Information systems environment: at least 18 additional SKS (or equivalent) of a cohesive set of topics that provide an understanding of an information systems environment.
- (c) Quantitative analysis or methods that must include statistics.

Discipline Criteria for Information Technology and Similarly-named Computing Programs

Lead Society(ies):

- *Asosiasi Pendidikan Tinggi Informatika dan Komputer (APTIKOM)* – Association of Higher Education Programs in Informatics and Computer

These program criteria apply to computing programs using information technology or similar terms in their titles.

Learning Outcomes 1.2.f.

In addition to graduate competencies of 1.2.a through 1.2.e required by the Common Criteria, graduates of the program will also have an ability to identify and analyze user needs and to take them into account in the selection, creation, integration, evaluation, and administration of computing based systems.

Curriculum

The curriculum requirements specify topics, but do not prescribe specific courses. The curriculum must include coverage of fundamentals and applied practice in the following:

- (a) The core information technologies of human-computer interaction, information management, programming, web systems and technologies, and networking.
- (b) System administration and system maintenance.
- (c) System integration and system architecture

Discipline Criteria for Software Engineering and Similarly-named Computing Programs

Lead Society(ies):

- *Asosiasi Pendidikan Tinggi Informatika dan Komputer (APTIKOM)* – Association of Higher Education Programs in Informatics and Computer

These program criteria apply to computing programs using software engineering or similar terms in their titles.

Learning Outcomes 1.2.f.

In addition to graduate competencies of 1.2.a through 1.2.e required by the Common Criteria, graduates of the program will also have an ability to develop a software system to meet desired needs within realistic constraints and resources.

Curriculum

The curriculum must provide both breadth and depth across the range of engineering and computer science topics implied by the title and objectives of the program.

The curriculum must include computing fundamentals, software design and construction, requirements analysis, security, verification, and validation; software engineering processes and tools appropriate for the development of complex software systems; and discrete mathematics, probability, and statistics, with applications appropriate to software engineering.